

The empirical recurrence for OEIS A281031, recovered from its tail

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Abstract

OEIS A281031 records “Empirical recurrence of order 98” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 1284$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 12$, so the recurrence is proved for every $n \geq 110$. Its last coefficient is $c_{98} = -64 \neq 0$, so no further tail satisfies anything shorter; any order-98 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A281031 is “Number of $n \times 5$ 0..1 arrays with no element equal to more than three of its king-move neighbors, with the exception of exactly one element, and with new values introduced in order 0 sequentially upwards.”. It has offset 1 and begins

0, 58, 2321, 9854, 44395, 184290, 722952, 2696692, ...

The entry states:

Empirical recurrence of order 98 (see link above)

As of the “Last modified” line on the live entry (Jan 13 2017 07:19:23, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 1284$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1284$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 98: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 12$ is the first whose minimal order is exactly the stated 98.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 2568$ exact terms from $n = 12$ on, it returns order 98 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{98} c_i a(n-i),$$

c_1	12	c_{26}	−2354722	c_{51}	6206820	c_{76}	2867948
c_2	−60	c_{27}	3149390	c_{52}	−3356410	c_{77}	2719602
c_3	166	c_{28}	−3206446	c_{53}	7906256	c_{78}	1972812
c_4	−276	c_{29}	2108306	c_{54}	−6638989	c_{79}	1104652
c_5	216	c_{30}	−11435	c_{55}	−4982224	c_{80}	246627
c_6	263	c_{31}	−3082610	c_{56}	−5929613	c_{81}	−323432
c_7	−1390	c_{32}	6092603	c_{57}	−532166	c_{82}	−534836
c_8	3444	c_{33}	−8209764	c_{58}	9117127	c_{83}	−532404
c_9	−6192	c_{34}	8744925	c_{59}	5578090	c_{84}	−429912
c_{10}	7720	c_{35}	−7118456	c_{60}	4872429	c_{85}	−248220
c_{11}	−4678	c_{36}	2565100	c_{61}	1633858	c_{86}	−104332
c_{12}	−6784	c_{37}	2721952	c_{62}	−4089518	c_{87}	−22712
c_{13}	28706	c_{38}	−8058337	c_{63}	−65580	c_{88}	15860
c_{14}	−60587	c_{39}	13371198	c_{64}	1158024	c_{89}	29672
c_{15}	92294	c_{40}	−14024424	c_{65}	1374910	c_{90}	22780
c_{16}	−100190	c_{41}	13172118	c_{66}	3551945	c_{91}	11104
c_{17}	53836	c_{42}	−8989583	c_{67}	1863814	c_{92}	4928
c_{18}	66784	c_{43}	1266974	c_{68}	−598019	c_{93}	1312
c_{19}	−266212	c_{44}	6589771	c_{69}	−3037390	c_{94}	−368
c_{20}	515993	c_{45}	−9859092	c_{70}	−4544652	c_{95}	−640
c_{21}	−706836	c_{46}	14679938	c_{71}	−5496114	c_{96}	−256
c_{22}	725889	c_{47}	−12590172	c_{72}	−4130108	c_{97}	−128
c_{23}	−430042	c_{48}	7704776	c_{73}	−1857174	c_{98}	−64
c_{24}	−266024	c_{49}	−8234994	c_{74}	153525		
c_{25}	1290964	c_{50}	−4270110	c_{75}	1926796		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 110$, and it is the recurrence OEIS A281031 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 12$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{98} - c_1x^{98-1} - \dots - c_{98}$. Its constant term is $-c_{98} = 64$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 98 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 98, so $\deg q = 98$, and it is monic. It holds from some index t on. From $\max(t, 12)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 98, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 22 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 98, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -64 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-98 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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